Hallucination Case Study
Assignment 3 — Smart Water Lab

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1 AI claim

During test design (Week 8 Session A, Cursor Agent): "For CN=95 and rainfall $P = 5$ mm, runoff is always zero because high CN means no runoff for small storms."

2 Why this is wrong

SCS-CN initial abstraction: $I_a = 0.2 \times (25400/\mathrm{CN} - 254)$.
For CN=95: $S \approx 13.37$ mm, $I_a \approx 2.67$ mm, so $P = 5$ mm gives $Q > 0$ — the verbal rule "high CN always means zero runoff for small storms" is wrong.

3 Tests created

`tests/test_runoff.py`: `test_runoff_below_initial_abstraction`, `test_runoff_not_exceed_rainfall_hig`
`src/validation.py`: `validate_runoff_mm` enforces $Q \leq P$.

4 Failure discovered

Draft documentation over-generalized one (CN, P) pair. pytest + Ia table showed the verbal rule was misleading for test design.

5 Fix applied

- Zero-runoff test: $P = 2$ mm, CN=80 ($I_a \approx 20.3$ mm).
- Physical sweep: CN = 60, 80, 95 at $P = 100$ mm with $Q \leq P$ assert.
- Documented in `prompt_log.md` and Week 8A lab report.

6 Evidence checklist

Step	Location
AI prompt	`prompt_log.md` Week 8A
Tests	`tests/test_runoff.py`
Failure documented	This file + Week 8A PDF
Fix	Corrected tests + validation layer
Run	`pytest -q` → 33 passed